Week 3a - Quiz

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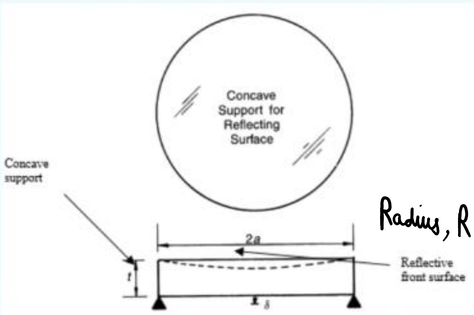
Problem 1 Mirrors for large telescopes - Introduction

There are some very large optical telescopes in the world. The newer ones employ complex and cunning tricks to maintain their precision as they track across the sky. The largest of these telescopes often contain a reflector that is a single large mirror, the largest of this kind being Subaru on Mount Kea in Hawaii. Here the mirror has a diameter of 8.3m and it is made of rigid glass; it is 20cm thick and weighs 22.8 tonnes. The average error in the surface profile is less than 0.014 microns. To stop the mirror from flexing under its own weight, active optics use 261 actuators to constantly flex the mirror to keep it in shape.

In the 20th century, telescopes were unactuated. The record holder for the world's largest mirror from 1975 until 1990 was on Mount Semivodrike, near Zelenchukskaya in the Caucasus Mountains of Russia. The mirror has a diameter of 6m, was also made of rigid glass and had a thickness of 1m and weighed more than 70 tonnes. These mirrors have to be stiff enough to position the mirror relative to the collecting system with precision about equal to that of the wavelength of light.

These large mirrors are made of glass with a silvered front surface (often a very thin aluminium layer) that is vapour deposited onto the glass. None of the optical properties of glass is used, therefore the glass was chosen purely for its mechanical properties. The mirror itself is made up of two parts: a reflective front surface and a concave support. Is there a better, lighter material available for a 6m diameter concave support?

The main objective for this question is to minimise the mass of the large light collecting mirror. A mirror is a circular disc with a diameter of $2a$ and a mean thickness of t, simply supported at its periphery (see figure). When horizontal it deflects under its own weight m, when vertical it does not deflect significantly. This distortion must be small enough that it does not interfere with performance. This means that the deflection, δ, of the midpoint of the mirror must be less than the wavelength of light. Assume that the density of the disk materials is ρ(rho).



Radius, $R = a$

Figure 1. The mirror of a large optical telescope

Calculate the mass of the mirror, which is the function that we will want to minimise?

m=

Display response

$$m = \rho V = \rho \times (\pi a^2 \times t) = \boxed{\pi a^2 \rho t}$$

The elastic deflection, $d\,(10\mu\text{m})$, is given by

$$\delta = \frac{3}{4\pi}\frac{mga^2}{Et^3} \qquad \text{(eq 2)}$$

Where g is acceleration due to gravity (9.81ms^{-2}) and E is Young's modulus.

Solving for t this gives:

$$t = \left(\frac{3}{4\pi}\right)^{\frac{1}{3}}\left(\frac{mga^2}{E\delta}\right)^{\frac{1}{3}} \qquad \text{(eq 3)}$$

$m = \pi a^2 \rho\, t$

Substituting equation (3) into the earlier mass equation gives:

$$m = \left(\frac{3g}{4\delta}\right)^{\frac{1}{2}}\left(\pi^{\frac{1}{4}}a\right)^{4}\left(\frac{\rho}{E^{\frac{1}{3}}}\right)^{\frac{3}{2}} \qquad \text{(eq 4)}$$

Equation (4) has been expressed in a form so as to find the performance index (M_1)

$$p = f_1(F).\,f_2(G).\,f_3(M) \qquad \text{(eq 5)}$$

In this final equation F is the functional requirements, G is the geometric parameters and M gives the material index.

From equation (4) it can be seen that weight is minimised by maximising which materials index?

$m\downarrow$ & $M\uparrow$

Select one:

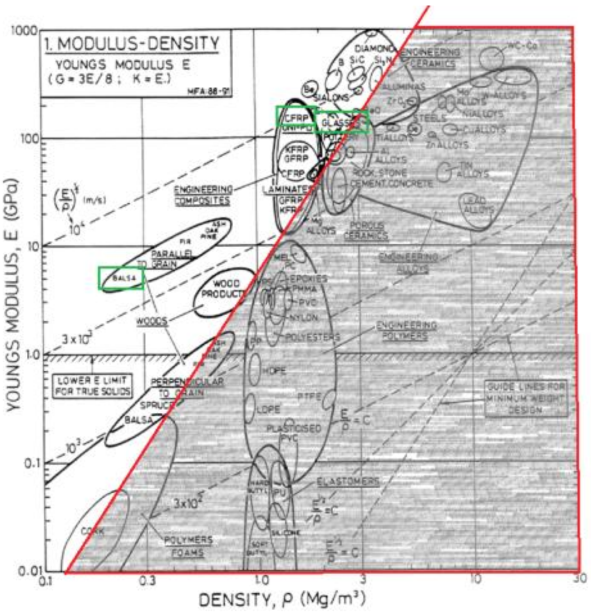
- a. $M_1 = \left(\frac{E^{\frac{1}{2}}}{\rho}\right)$ ×
- b. $M_1 = \left(\frac{E^{\frac{1}{3}}}{\rho}\right)$ ✓
- c. $M_1 = \left(\frac{E}{\rho}\right)$ ×
- d. $M_1 = \left(\frac{\rho}{E}\right)$ ×

$$m = \left(\frac{3g}{4\delta}\right)^{\frac{1}{2}}\left(\pi^{\frac{1}{4}}a\right)^{4}\left(\frac{\rho}{E^{1/3}}\right)^{3/2}$$

$$\rho = f_1(F)\, f_2(G)\, f_3(M)$$

$$\boxed{\therefore M = \frac{1}{\left(\frac{\rho}{E^{1/3}}\right)} = \frac{E^{1/3}}{\rho}}$$

Position the selection line until you identify three sensible alternative materials for the reflecting mirror. Highlight your materials of choice.



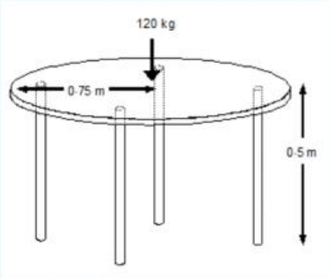
Explain what is your optimum material and give reasons for your choice.

Glass is the optimum material; CFRP is too expensive, and balsa is too weak, and both of those are unable to have vapour deposition of aluminium.

Problem 2 - Axially loaded slender table legs.

A specialist furniture designer who wishes to build an extremely simple lightweight table. He would like the table to be constructed from a flat sheet of toughened glass simply supported by four slender, unbraced, cylindrical legs (figure 4). The table will stand 0.5 m off the ground. The designer wants the table to look like it is floating in the air by having what appear to be legs that are too thin to support the table's weight.

The designer wishes that the legs are solid (such that they can be made thinner) and also must be as lightweight as possible (so that it can be easily moved), but the legs should not be susceptible to breakage in everyday use. Furthermore, the legs must be able to support the glass table top and anything placed upon it without buckling.



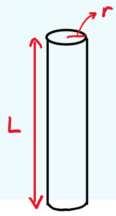
The maximum perceived additional load that the table will take is 120 kg. The glass table top will be made from a sheet 8.5 mm thick and will be 1.5 m in diameter. The selected glass has a density of 2.3264 Mgm^{-3}, with the cost estimated to be £2/kg.

The designer would like a comparison between wood and metals, and more modern materials such as composites. Above all, in order to sell as many as possible and maximise his profits, it is essential that the total cost of the materials for the tables be no more than £200.

this problem has multiple design goals as we aim to minimise the overall weight and make the legs as thin as possible. There are some constraints that we want to avoid the leg breaking particularly as a consequence of buckling.

If we consider each leg to be a slender column of material of a density ρ and modulus E. The length, l, and the maximum load, P, that the table can carry are fixed by the design. Only the leg radius r is a free variable.

Calculate the mass of one table leg, which is the first function that we will want to minimise?



m=

Display response

$m = \rho V = \rho \times \pi r^2 L = \rho \pi r^2 l$

The elastic buckling load P_{crit} of a column of a length l and radius r is given by

$P_{crit} = \frac{\pi^2 EI}{l^2}$ (eq 2)

where the second moment of inertia for a column is

$I = \pi r^4/4$ (eq 3)

Knowing that the load P must not exceed P_{crit} the solution for the point of buckling for the leg is given as:

$P_{crit} = \frac{\pi^3 E r^4}{4l^2}$ (eq 4)

solving for the free variable r and substituting it into the mass equation for the leg gives

$m \geq (\frac{4P}{\pi})^{1/2}(l^2)(\frac{\rho}{E^{1/2}})$ (eq 5)

From equation (5) it can be seen that weight is minimised by maximising which materials index?

$\downarrow$ $m\downarrow$

$\downarrow$ $\left(\frac{\rho}{E^{1/2}}\right)\downarrow = \left(\frac{E^{1/2}}{\rho}\right)\uparrow$

Select one:

- a. $M_1 = (\frac{E^{\frac{1}{2}}}{\rho})$ ✓
- b. $M_1 = (\frac{E}{\rho})$ ×
- c. $M_1 = (\frac{\rho}{E})$ ×
- d. $M_1 = (\frac{E^{\frac{1}{3}}}{\rho})$ ×

The next design criterion is that of slenderness. From equation (2), derive an equation for slenderness assuming that $P \leq P_{crit}$, and calculate the next performance index (M_2).

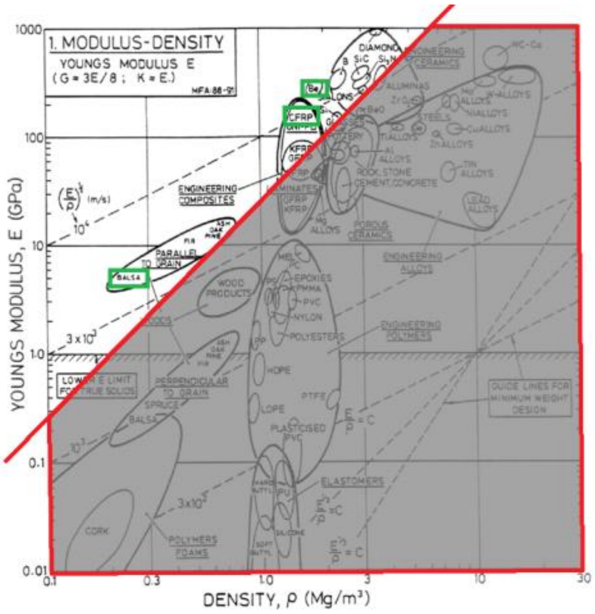
$$P_{crit} = \frac{\pi^2 E I}{l^2}$$

$$\therefore P \leq P_{crit} = \left(\frac{\pi^2 I}{l^2}\right) \cdot E$$

Select one:

- a. $M_2 = (\frac{\rho}{E})$ ✗
- b. $M_2 = (\frac{E}{\rho})$ ✗
- c. $M_2 = E$ ✓
- d. $M_2 = (\frac{E^{\frac{1}{2}}}{\rho})$ ✗

Position the selection line until you identify three sensible alternative materials for the table leg. Highlight your materials of choice.



- CFRP is expensive, but allows for r to be the smallest
- Balsa is cheap, but not dense or strong enough to warrant the small value of r.
- Be is toxic.